

# London Airbnb Investment Analysis

## End-to-End Data Analytics Case Study

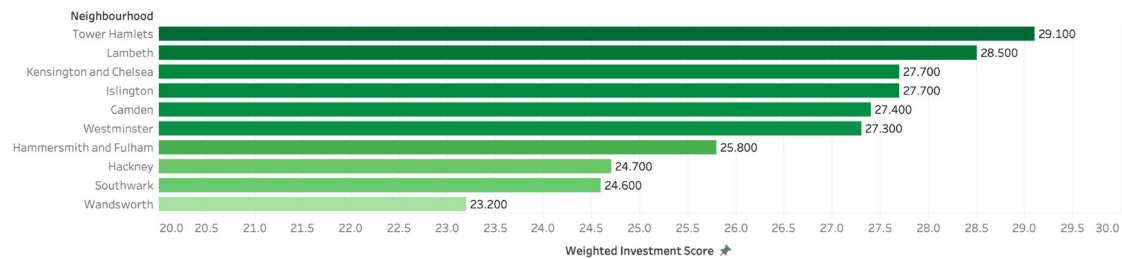
Prepared by: Afridi Islam

Tools: Google Sheets, Google BigQuery, SQL, Tableau

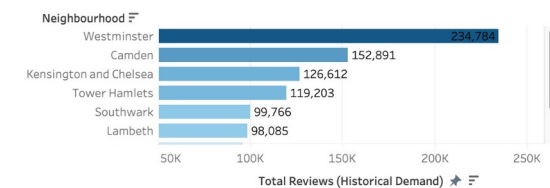
Dataset: Inside Airbnb - London, England, United Kingdom - 14 September 2025

London Airbnb Investment Analysis Dashboard

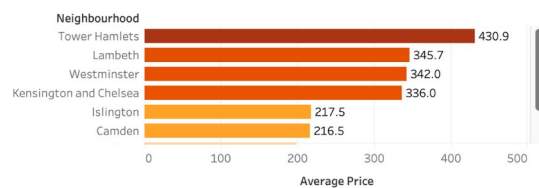
Top 10 Investment Neighbourhoods in London



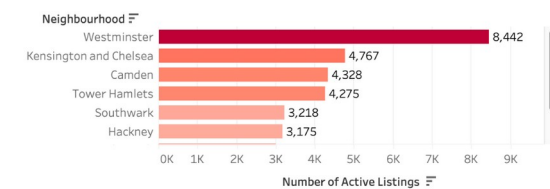
Historical Demand by Neighbourhood



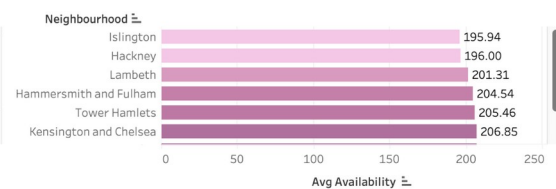
Average Nightly Price



Market Competition by Neighbourhood



Average Listing Availability



Final Tableau dashboard created during the Share phase

### Project Phases

| Phase   | Purpose                                                                         | Status    |
|---------|---------------------------------------------------------------------------------|-----------|
| Ask     | Define the business task, stakeholders, business questions, and success metrics | Completed |
| Prepare | Assess datasets, credibility,                                                   | Completed |

|         |                                                               |           |
|---------|---------------------------------------------------------------|-----------|
|         | structure, and scope                                          |           |
| Process | Clean, validate, and create analysis-ready tables             | Completed |
| Analyze | Answer business questions and build weighted investment model | Completed |
| Share   | Communicate insights through Tableau dashboard                | Completed |
| Act     | Recommend business actions, limitations, and future work      | Completed |

# Executive Summary

This project analyzed Airbnb listings across London to identify the strongest neighbourhoods for short-term rental investment. The analysis followed the Google Data Analytics process: Ask, Prepare, Process, Analyze, Share, and Act.

The analysis used London Airbnb data from Inside Airbnb, focusing on listing-level metrics such as price, reviews, room type, availability, and neighbourhood. Google Sheets was used for initial inspection, Google BigQuery and SQL were used for cleaning and analysis, and Tableau was used for dashboard development.

A weighted investment scoring model was built using four business metrics: historical demand, average nightly price, market competition, and average availability. The model identified Tower Hamlets, Lambeth, and Kensington & Chelsea as the strongest overall investment opportunities.

| Final Recommendation | Reason                                                                                                |
|----------------------|-------------------------------------------------------------------------------------------------------|
| Tower Hamlets        | Highest investment score, highest average price, strong demand, and favorable availability indicator. |
| Lambeth              | Strong overall balance across price, demand, competition, and availability.                           |
| Kensington & Chelsea | High demand and strong pricing despite higher competition.                                            |

## 1. Ask Phase

The Ask phase defined the business problem, stakeholders, business questions, and success metrics before any cleaning or analysis was performed.

### Business Scenario

A fictional real estate management company wants to expand into the London short-term rental market. The company has enough capital to invest in only three London neighbourhoods and needs evidence-based recommendations before purchasing properties.

### Business Task

Analyze the London Airbnb market to identify neighbourhoods with strong demand, competitive pricing, manageable competition, and high investment potential, enabling the client to make informed property acquisition decisions.

### Stakeholders

| Stakeholder Group | Stakeholders                                                     |
|-------------------|------------------------------------------------------------------|
| Primary           | CEO, Investment Manager, Property Acquisition Team, Finance Team |
| Secondary         | Marketing Team, Property Managers,                               |

|  |                 |
|--|-----------------|
|  | Operations Team |
|--|-----------------|

## Business Questions

1. Which London neighbourhoods offer the best Airbnb investment opportunities?
2. Which neighbourhoods show the strongest customer demand?
3. Which neighbourhoods have the highest revenue potential based on average nightly price?
4. Which neighbourhoods have the most market competition?
5. Which neighbourhoods show lower availability, potentially indicating stronger occupancy?
6. Which room type performs best for Airbnb investment?

## Success Metrics

| Business Objective      | Metric                        | Dataset Column(s)                    |
|-------------------------|-------------------------------|--------------------------------------|
| Demand                  | Historical demand             | number_of_reviews, reviews_per_month |
| Revenue potential       | Average nightly price         | price                                |
| Competition             | Number of listings            | COUNT(*) by neighbourhood            |
| Risk / occupancy signal | Average annual availability   | availability_365                     |
| Room type strategy      | Demand and price by room type | room_type, number_of_reviews, price  |

## 2. Prepare Phase

The Prepare phase focused on understanding the data source, evaluating dataset credibility, selecting the appropriate files, and planning the cleaning strategy.

### Data Source

The data was obtained from the Inside Airbnb project for London, England, United Kingdom, published on 14 September 2025. The dataset contains listing, review, neighbourhood, and calendar files.

| Dataset            | Rows / Columns             | Use in Project   | Reason                                                                                 |
|--------------------|----------------------------|------------------|----------------------------------------------------------------------------------------|
| listings.csv       | 96,872 rows / 18 columns   | Yes              | Primary dataset containing neighbourhood, room type, price, reviews, and availability. |
| reviews.csv        | 2,097,997 rows / 2 columns | Yes / supporting | Useful for review activity and demand context.                                         |
| neighbourhoods.csv | 34 rows / 2 columns        | Reference        | Used as a reference table for neighbourhood names.                                     |

|                 |                 |                             |                                                                                 |
|-----------------|-----------------|-----------------------------|---------------------------------------------------------------------------------|
| calendar.csv.gz | Compressed file | Excluded from initial scope | Not inspected in detail; primary business questions were answerable without it. |
|-----------------|-----------------|-----------------------------|---------------------------------------------------------------------------------|

## ROCCC Assessment

| Criterion     | Assessment                                                                                               |
|---------------|----------------------------------------------------------------------------------------------------------|
| Reliable      | Publicly available Airbnb listing data; suitable for educational and exploratory analysis.               |
| Original      | Published by Inside Airbnb, which gathers publicly available Airbnb data.                                |
| Comprehensive | Includes listing price, room type, neighbourhood, reviews, availability, and location-related variables. |
| Current       | Dataset snapshot published on 14 September 2025.                                                         |
| Cited         | Source can be documented through the Inside Airbnb website.                                              |

## Important Columns Identified

| Column Category               | Columns                                                                                                     |
|-------------------------------|-------------------------------------------------------------------------------------------------------------|
| Essential                     | id, neighbourhood, room_type, price, minimum_nights, availability_365, number_of_reviews, reviews_per_month |
| Useful                        | number_of_reviews_ltm, calculated_host_listings_count, latitude, longitude, name, last_review               |
| Unnecessary for this analysis | license, host_name, host_id, neighbourhood_group (empty)                                                    |

## Planned Quality Checks

- Check for duplicate records.
- Check missing values in key fields.
- Check inconsistent formats and data types.
- Check extra spaces and text inconsistencies where relevant.
- Validate numeric fields using business rules.
- Identify extreme or unrealistic values such as price  $\leq 0$  or availability  $> 365$ .

### 3. Process Phase

The Process phase cleaned and validated the dataset. Google Sheets was used for initial inspection, while BigQuery was used for reproducible cleaning and validation due to the dataset size.

#### Importing Data into BigQuery

The listings CSV initially produced import errors due to malformed CSV rows and quote parsing issues in listing names. The issue was resolved by adjusting BigQuery CSV import settings, including allowing quoted newlines, jagged rows, unknown values, and increasing the maximum bad records threshold.

A small number of malformed rows may have been skipped during import. This was documented as a limitation, and the successfully imported BigQuery table was treated as the source of truth for subsequent cleaning.

#### Missing Value Handling

- Some missing values were found in `host_name`. These were retained because `host_name` was not required for the investment analysis.
- The `neighbourhood_group` column was completely empty and was excluded from the cleaned dataset.
- The price column contained 34,903 missing values, approximately 36% of the imported dataset.
- Rows with missing price values were removed because price is essential for revenue and investment analysis.

#### Creating the Clean Table

```
CREATE TABLE `airbnb-case-study-london-2025.airbnb_london_analysis.airbnb_london_clean` AS
SELECT *
FROM `airbnb-case-study-london-2025.airbnb_london_analysis.airbnb_london`
WHERE price IS NOT NULL;
```

After cleaning, the table `airbnb_london_clean` contained 61,945 records.

#### Validation Queries

The following business-rule validation checks were performed:

| Validation Check            | SQL Condition                                                         | Result           |
|-----------------------------|-----------------------------------------------------------------------|------------------|
| Missing price removed       | <code>price IS NULL</code>                                            | 0 in clean table |
| Invalid price               | <code>price &lt;= 0</code>                                            | 0                |
| Invalid minimum nights      | <code>minimum_nights &lt;= 0</code>                                   | 0                |
| Invalid availability        | <code>availability_365 &lt; 0 OR<br/>availability_365 &gt; 365</code> | 0                |
| Invalid reviews per month   | <code>reviews_per_month &lt; 0</code>                                 | 0                |
| Invalid number of reviews   | <code>number_of_reviews &lt; 0</code>                                 | 0                |
| Invalid host listings count | <code>calculated_host_listings_count<br/>&lt; 0</code>                | 0                |

```
SELECT COUNT(*) AS invalid_availability
FROM `airbnb-case-study-london-2025.airbnb_london_analysis.airbnb_london_clean`
WHERE availability_365 < 0
      OR availability_365 > 365;
```

The cleaned dataset was considered reliable enough for the business questions because all essential numeric fields passed validation.

## 4. Analyze Phase

The Analyze phase answered the business questions using SQL in BigQuery. The analysis moved from simple descriptive metrics to a weighted investment scoring model.

### 4.1 Historical Demand by Neighbourhood

Historical demand was measured using total guest reviews by neighbourhood.

```
SELECT
    neighbourhood,
    SUM(number_of_reviews) AS total_reviews
FROM `airbnb-case-study-london-2025.airbnb_london_analysis.airbnb_london_clean`
GROUP BY neighbourhood
ORDER BY total_reviews DESC
LIMIT 10;
```

| Rank | Neighbourhood          | Total Reviews |
|------|------------------------|---------------|
| 1    | Westminster            | 234,784       |
| 2    | Camden                 | 152,891       |
| 3    | Kensington and Chelsea | 126,612       |
| 4    | Tower Hamlets          | 119,203       |
| 5    | Southwark              | 99,766        |
| 6    | Lambeth                | 98,085        |
| 7    | Islington              | 95,870        |
| 8    | Hackney                | 92,182        |
| 9    | Hammersmith and Fulham | 78,001        |
| 10   | Wandsworth             | 73,015        |

Insight: Westminster recorded the strongest historical demand, followed by Camden and Kensington & Chelsea.

### 4.2 Average Price by Neighbourhood

```
SELECT
    neighbourhood,
    AVG(price) AS avg_price
FROM `airbnb-case-study-london-2025.airbnb_london_analysis.airbnb_london_clean`
GROUP BY neighbourhood
ORDER BY avg_price DESC
LIMIT 10;
```

| Rank | Neighbourhood | Average Price |
|------|---------------|---------------|
| 1    | Tower Hamlets | 430.91        |

|    |                        |        |
|----|------------------------|--------|
| 2  | City of London         | 354.06 |
| 3  | Lambeth                | 345.69 |
| 4  | Westminster            | 342.05 |
| 5  | Kensington and Chelsea | 335.98 |
| 6  | Islington              | 217.52 |
| 7  | Camden                 | 216.54 |
| 8  | Hammersmith and Fulham | 199.16 |
| 9  | Wandsworth             | 198.41 |
| 10 | Richmond upon Thames   | 184.27 |

Insight: The neighbourhoods with the highest demand did not always have the highest average prices. Tower Hamlets ranked first by average price while Westminster ranked first by historical demand.

### 4.3 Competition by Neighbourhood

Competition was measured by counting the number of listings in each neighbourhood.

```
SELECT
    neighbourhood,
    COUNT(*) AS total_listings
FROM `airbnb-case-study-london-2025.airbnb_london_analysis.airbnb_london_clean`
GROUP BY neighbourhood
ORDER BY total_listings DESC
LIMIT 10;
```

| Rank | Neighbourhood          | Listings |
|------|------------------------|----------|
| 1    | Westminster            | 8,442    |
| 2    | Kensington and Chelsea | 4,767    |
| 3    | Camden                 | 4,328    |
| 4    | Tower Hamlets          | 4,275    |
| 5    | Southwark              | 3,218    |
| 6    | Hackney                | 3,175    |
| 7    | Wandsworth             | 3,011    |
| 8    | Lambeth                | 2,819    |
| 9    | Islington              | 2,755    |
| 10   | Hammersmith and Fulham | 2,668    |

Insight: Westminster had the highest competition, which reduces its overall investment attractiveness despite strong demand.

### 4.4 Availability by Neighbourhood

Average availability was used as an indirect occupancy indicator. Lower availability may indicate stronger booking activity.

```
SELECT
    neighbourhood,
    AVG(availability_365) AS avg_availability
FROM `airbnb-case-study-london-2025.airbnb_london_analysis.airbnb_london_clean`
```



```
GROUP BY neighbourhood
ORDER BY avg_availability ASC
LIMIT 10;
```

| Rank | Neighbourhood          | Average Availability |
|------|------------------------|----------------------|
| 1    | Islington              | 195.94               |
| 2    | Hackney                | 196.00               |
| 3    | Lambeth                | 201.31               |
| 4    | Hammersmith and Fulham | 204.54               |
| 5    | Tower Hamlets          | 205.46               |
| 6    | Richmond upon Thames   | 205.83               |
| 7    | Kensington and Chelsea | 206.85               |
| 8    | Camden                 | 207.12               |
| 9    | Brent                  | 207.23               |
| 10   | Haringey               | 207.51               |

Insight: Islington and Hackney had the lowest average availability, while Lambeth and Tower Hamlets also performed well on this metric.

### 4.5 Room Type Analysis

Room type analysis was performed to determine what type of property investors should prioritize.

| Room Type       | Historical Demand | Average Price |
|-----------------|-------------------|---------------|
| Entire home/apt | 910,459           | 279.33        |
| Private room    | 748,093           | 121.70        |
| Shared room     | 7,273             | 96.91         |
| Hotel room      | 3,413             | 657.83        |

Insight: Hotel rooms had the highest average price, but very low demand. Entire homes/apartments combined the highest historical demand with strong average prices, making them the most attractive room type for investment.

### 4.6 Weighted Investment Scoring Model

A weighted scoring model was built to evaluate neighbourhoods across multiple metrics rather than choosing based on a single factor. CTEs and SQL window functions were used to aggregate metrics, rank neighbourhoods, and calculate a final investment score.

| Metric            | Weight | Ranking Direction | Reason                                            |
|-------------------|--------|-------------------|---------------------------------------------------|
| Historical Demand | 40%    | Higher is better  | More reviews indicate stronger historical demand. |
| Average Price     | 30%    | Higher is better  | Higher nightly prices increase revenue potential. |
| Availability      | 20%    | Lower is better   | Lower availability may indicate higher            |

|             |     |                 |                                              |
|-------------|-----|-----------------|----------------------------------------------|
|             |     |                 | occupancy.                                   |
| Competition | 10% | Lower is better | Fewer listings may make market entry easier. |

```

WITH summary_table AS (
    SELECT
        neighbourhood,
        SUM(number_of_reviews) AS demand,
        AVG(price) AS avg_price,
        COUNT(*) AS total_listings,
        AVG(availability_365) AS avg_availability
    FROM `airbnb-case-study-london-2025.airbnb_london_analysis.airbnb_london_clean`
    GROUP BY neighbourhood
),

ranked_table AS (
    SELECT
        *,
        RANK() OVER (ORDER BY demand DESC) AS demand_rank,
        RANK() OVER (ORDER BY avg_price DESC) AS price_rank,
        RANK() OVER (ORDER BY total_listings ASC) AS competition_rank,
        RANK() OVER (ORDER BY avg_availability ASC) AS availability_rank
    FROM summary_table
)

SELECT
    neighbourhood,
    demand,
    avg_price,
    total_listings,
    avg_availability,
    demand_rank,
    price_rank,
    competition_rank,
    availability_rank,
    ROUND(
        ((35 - demand_rank) * 0.40) +
        ((35 - price_rank) * 0.30) +
        ((35 - competition_rank) * 0.10) +
        ((35 - availability_rank) * 0.20),
        2
    ) AS investment_score
FROM ranked_table
ORDER BY investment_score DESC;

```

| Rank | Neighbourhood          | Investment Score |
|------|------------------------|------------------|
| 1    | Tower Hamlets          | 29.1             |
| 2    | Lambeth                | 28.5             |
| 3    | Kensington and Chelsea | 27.7             |
| 4    | Islington              | 27.7             |
| 5    | Camden                 | 27.4             |
| 6    | Westminster            | 27.3             |
| 7    | Hammersmith and Fulham | 25.8             |
| 8    | Hackney                | 24.7             |
| 9    | Southwark              | 24.6             |

|    |            |      |
|----|------------|------|
| 10 | Wandsworth | 23.2 |
|----|------------|------|

Insight: Tower Hamlets ranked highest overall. Lambeth performed strongly because it balanced price, demand, availability, and competition. Westminster had the highest historical demand but ranked lower overall because of high competition and weaker availability ranking.

## 5. Share Phase

The Share phase communicated the findings through an executive Tableau dashboard. The dashboard was designed to answer the question: Which London neighbourhoods provide the best Airbnb investment opportunities?

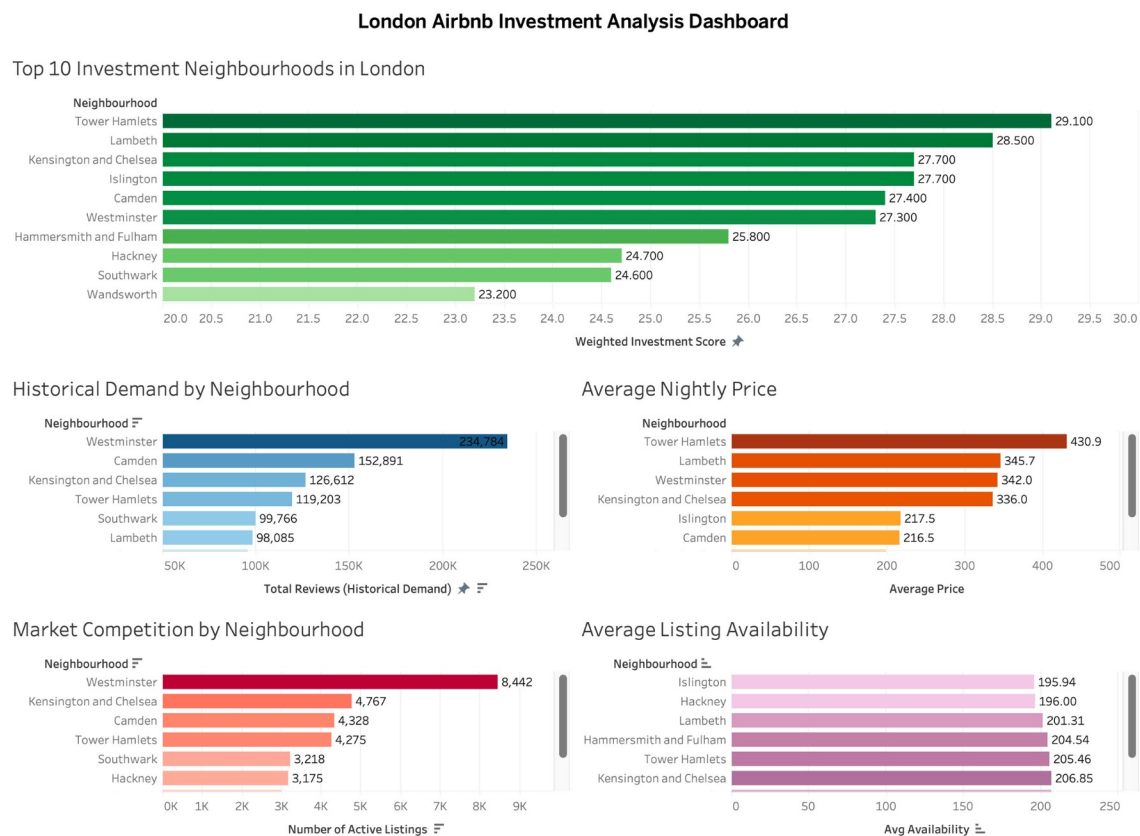


Figure: Final Tableau dashboard showing investment score, demand, price, competition, and availability.

## Dashboard Components

| Visualization                      | Business Question Answered                    |
|------------------------------------|-----------------------------------------------|
| Top Investment Neighbourhoods      | Where should investors prioritize investment? |
| Historical Demand by Neighbourhood | Where is customer demand strongest?           |
| Average Nightly Price              | Where is revenue potential highest?           |
| Market Competition                 | Which markets are most saturated?             |

|                      |                                   |
|----------------------|-----------------------------------|
| Average Availability | Where may occupancy be strongest? |
|----------------------|-----------------------------------|

## Design Decisions

- Horizontal bar charts were used because London neighbourhood names are long.
- Investment score was placed at the top because it answers the primary executive question.
- Each metric was assigned a consistent color: green for investment score, blue for demand, orange for price, red for competition, and purple for availability.
- Data labels were shown to make values readable without additional interaction.
- The supporting charts were placed below the recommendation chart to explain why the top neighbourhoods ranked highly.

## Key Dashboard Insights

- Tower Hamlets achieved the highest weighted investment score.
- Westminster had the highest historical demand but also the highest market competition.
- Lambeth and Kensington & Chelsea provided strong overall investment potential.
- Entire homes/apartments were identified as the strongest room type due to demand and price performance.

## 6. Act Phase

The Act phase translated the analysis into practical recommendations that stakeholders can use to guide investment decisions.

### Business Recommendations

7. Prioritize investment in Tower Hamlets, Lambeth, and Kensington & Chelsea because these neighbourhoods achieved the highest weighted investment scores.
8. Focus on Entire Home/Apartment listings because they demonstrated the strongest combination of historical demand and average nightly price.
9. Do not choose investment locations based solely on demand. Westminster had the highest historical demand, but its market competition reduced its overall attractiveness.
10. Use availability alongside demand and price because lower average availability may indicate stronger booking activity.
11. Continue using a multi-metric scoring framework to compare future investment opportunities objectively.

### Expected Business Impact

- Improved investment decision-making using objective, data-driven analysis.
- Higher revenue potential by targeting neighbourhoods with balanced market performance.
- Reduced risk by avoiding highly saturated markets.
- A repeatable framework for evaluating future Airbnb investment markets.

## Limitations

- The analysis used historical Airbnb listing data and may not represent future market conditions.
- Availability was used as a proxy for occupancy and does not represent confirmed bookings.
- Property purchase prices, operating costs, taxes, and maintenance expenses were not included.
- The scoring model depends on selected weights; different business priorities could justify different weights.
- Some malformed CSV records were skipped during BigQuery import due to source file formatting issues.

## Future Work

- Incorporate property acquisition costs and operating expenses.
- Calculate estimated annual revenue and true ROI.
- Analyze seasonality using calendar and review date data.
- Include guest ratings and review sentiment if available.
- Build geographic maps of investment opportunities.
- Develop predictive models for future demand and pricing trends.

## Final Conclusion

This project demonstrated an end-to-end data analytics workflow. The analysis moved from business problem definition to data preparation, cleaning, SQL analysis, Tableau visualization, and final recommendations. The final recommendation is to prioritize Tower Hamlets, Lambeth, and Kensington & Chelsea for London Airbnb investment, with a preference for Entire Home/Apartment properties.

## Appendix: Skills Demonstrated

| Category                 | Skills Demonstrated                                                                                 |
|--------------------------|-----------------------------------------------------------------------------------------------------|
| Spreadsheet / Inspection | Row counts, column review, missing value checks, initial data quality assessment.                   |
| SQL / BigQuery           | COUNT, SUM, AVG, GROUP BY, ORDER BY, WHERE, CTEs, RANK window functions, weighted scoring model.    |
| Data Cleaning            | Missing value handling, business-rule validation, clean table creation, preserving raw data.        |
| Analytics                | Demand analysis, pricing analysis, competition analysis, availability analysis, room type analysis. |
| Visualization            | Tableau dashboards, horizontal bar charts, metric-specific color strategy, executive layout.        |
| Business Communication   | Recommendations, limitations, future work, final conclusion.                                        |